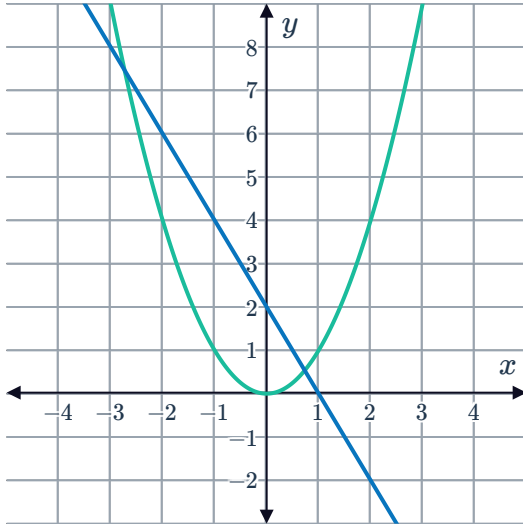


Points of intersection

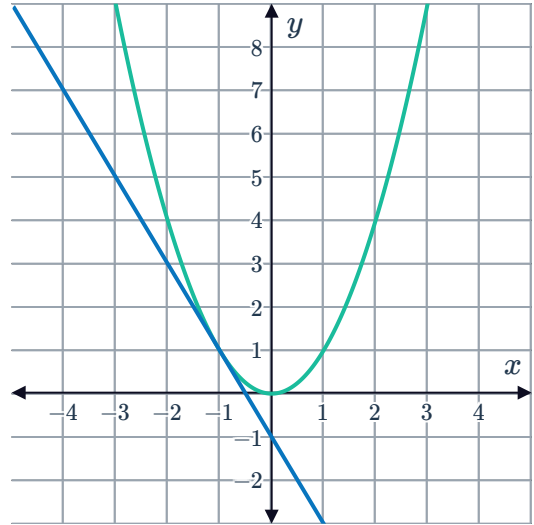


1 State the number of points of intersection for each graph:

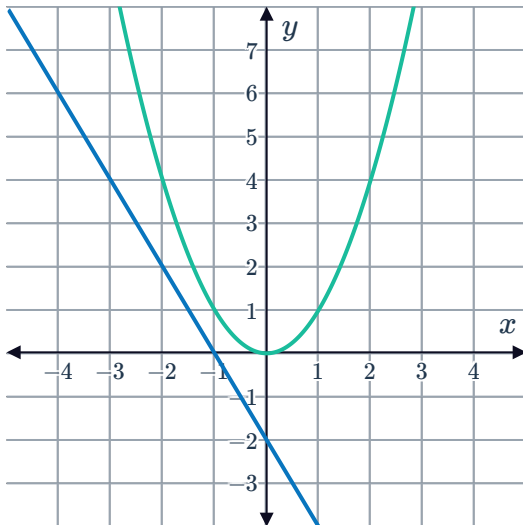
a



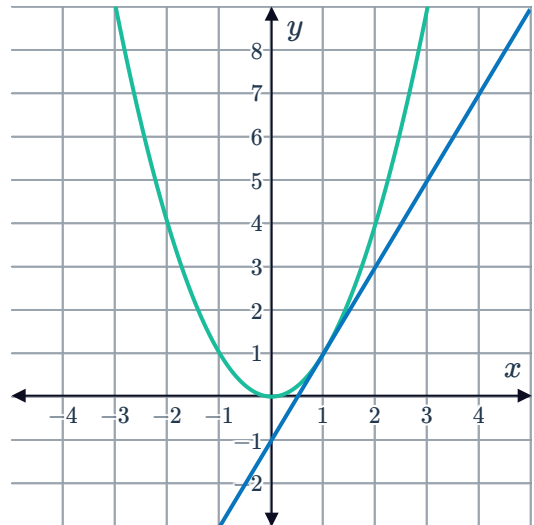
b



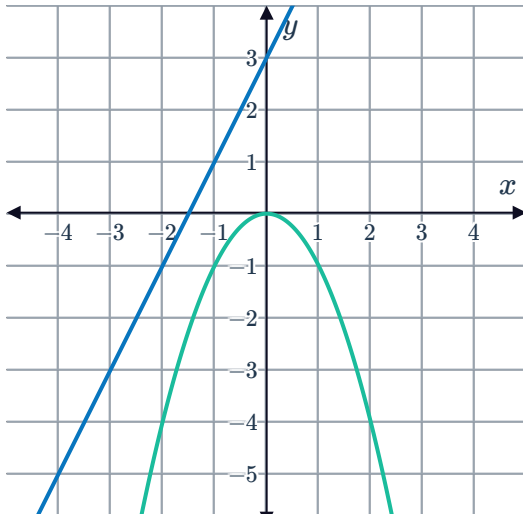
c



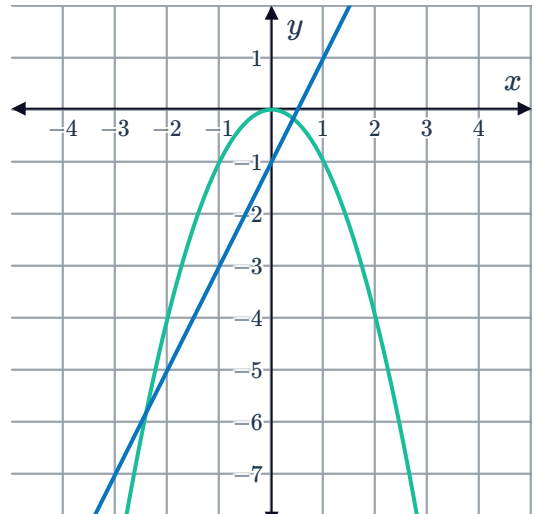
d



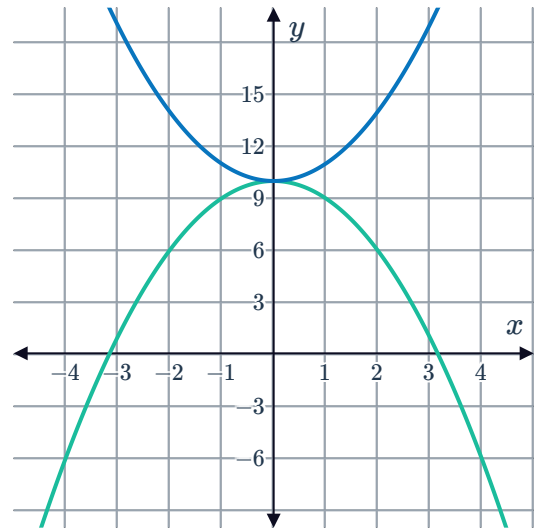
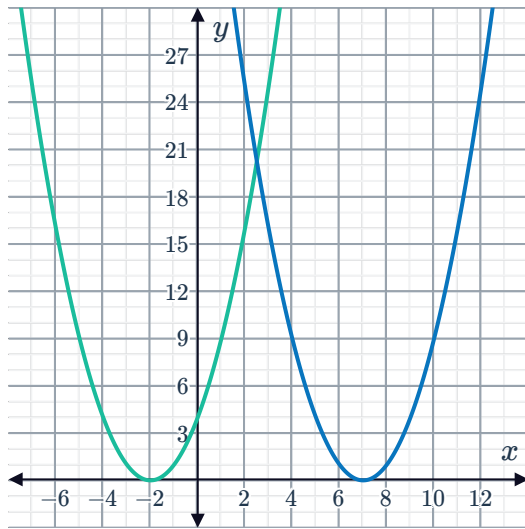
e



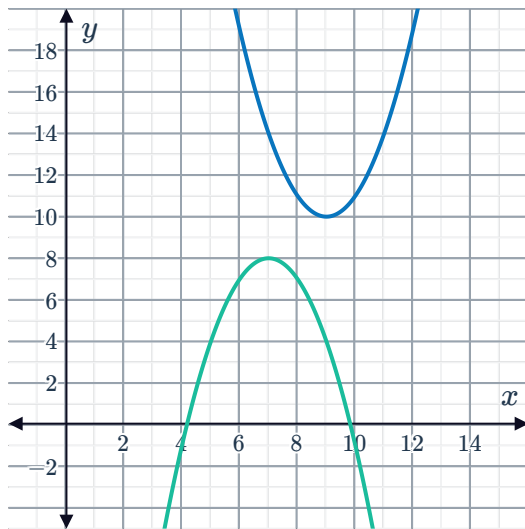
f



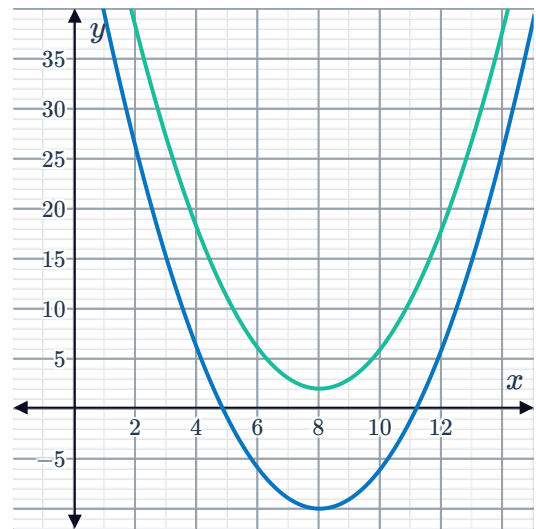
g



i



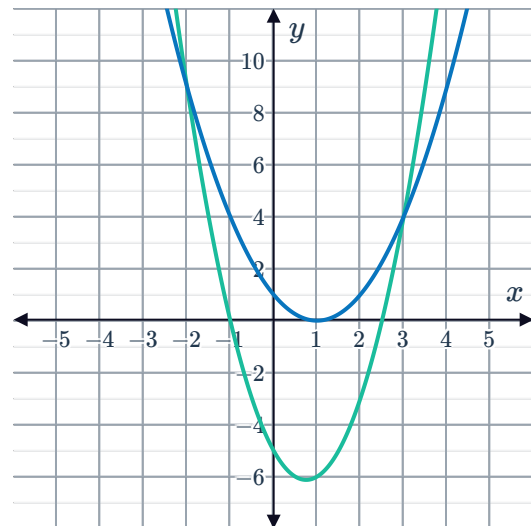
j



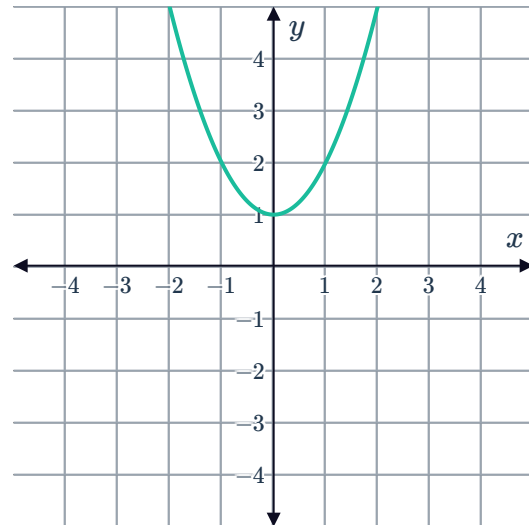
2 Consider the system of equations:

$$\begin{cases} y = 2x^2 - 3x - 5 \\ y = x^2 - 2x + 1 \end{cases}$$

What are the coordinates of the points of intersection?



- 3 The function $y = x^2 + 1$ is shown on the graph. State the minimum value of k such that the line $y = k$ has at least one point of intersection with the parabola.



- 4 Consider the following system of equations:
- $$\begin{cases} y = 4x^2 + 16 \\ y = 5x + 2 \end{cases}$$
- Use the two equations to form a quadratic equation in terms of x only.
 - Find the discriminant of the newly formed equation.
 - Do the equations have any common coordinates? Explain your answer.
- 5 When solving $y = x^2 - 3x + 1$ and $y = x + 6$ simultaneously, one point of intersection is at $x = -1$. What is the y -coordinate at this point?
- 6 Find the coordinates of the points where the vertical line $x = -5$ intersects the curve $y = -2x^2 + x - 12$.
- 7 One point of intersection of the curves $y = x^2 - 2x + 30$ and $y = kx + 10$ occurs at $x = 4$. Find the value of k .
- 8 For each of the following pairs of equations:
- Graph the equations on the same coordinate plane.
 - State the coordinates of the points of intersection.
- $$\begin{cases} y = x^2 + 4x + 1 \\ y = -x^2 + 7 \end{cases}$$
 - $$\begin{cases} y = -3x + 4 \\ y = 8 - x^2 \end{cases}$$

- 9 Consider the following system of equations 
$$\begin{cases} y = -4x^2 + 10x \\ y = c \end{cases}$$

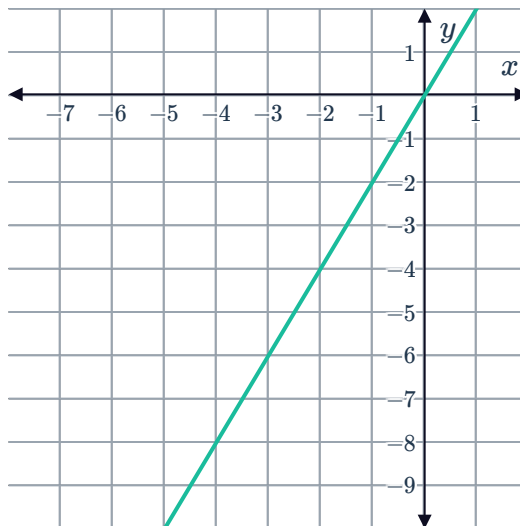
- Find the x -intercepts of the quadratic equation.
- Find the axis of symmetry of the graph corresponding to the quadratic equation.
- Now, determine the coordinates of the vertex for the quadratic equation.
- Graph the quadratic equation on the coordinate plane.
- For what value of c will there be one solution to the system of equations?
- For what values of c will there be two solutions to the system of equations?

- 10 Consider the equations $y = x^2 + 6x$ and $y = 2x$.

- a Complete the table of values for the two equations.

x	-4	-3	-2
$y = x^2 + 6x$			
$y = 2x$			

- b The line $y = 2x$ has been graphed. On the same coordinate plane, graph $y = x^2 + 6x$.



- c Now state the coordinates of the points of intersection.

11 Consider the equations $y = 4x - x^2$ and $y = \text{[Image]} + 5$.



a Complete the table of values for the two equations.

x	1	2	3
$y = 4x - x^2$			
$y = -2x + 5$			

b Graph $y = 4x - x^2$ and $y = -2x + 5$ on the same coordinate plane.

c Now state the coordinates of the points of intersection.

12 Consider the parabola $y = x^2 - 3x - 3$ and the line $x = -2$.

a If there is only one point of intersection between the two graphs, does this mean $x = -2$ is a tangent to the parabola?

b State the coordinates of the point of intersection.

13 Vanessa is using algebra to find the points of intersection of a parabola and a straight line. She finds a quadratic equation in the form $ax^2 + bx + c = 0$, but when she tries to solve it, she finds that the equation has no solutions. Graph an example of a parabola and a straight line that would satisfy this situation.

14 For each of the following pairs of equations:

i Sketch the graphs of the two equations on the same coordinate plane.

ii State the coordinates of the points of intersection.

a
$$\begin{cases} y = x^2 \\ y = 2x^2 - 1 \end{cases}$$

b
$$\begin{cases} y = x^2 + 1 \\ y = 3 - x^2 \end{cases}$$



Quadratic systems of equations

15 For each of the following pairs of equations:

- i Solve for the x -coordinates of the points of intersection.
- ii State the coordinates of the points of intersection.

a
$$\begin{cases} y = x^2 - 4 \\ y = 4 - x^2 \end{cases}$$

b
$$\begin{cases} y = x^2 \\ y = 4x^2 - 12 \end{cases}$$

c
$$\begin{cases} y = x^2 \\ y = x^2 + 4x \end{cases}$$

d
$$\begin{cases} y = (x + 2)^2 \\ y = 34 - x^2 \end{cases}$$

e
$$\begin{cases} y = x^2 - 7x + 8 \\ y = x^2 + 8x - 97 \end{cases}$$

f
$$\begin{cases} y = 2x^2 - 2x + 14 \\ y = x^2 + 7x - 6 \end{cases}$$

g
$$\begin{cases} y = 11x^2 - 10x - 5 \\ y = -5x^2 - 2x - 6 \end{cases}$$

h
$$\begin{cases} y = (x - 8)^2 - 5 \\ y = (x - 9)^2 - 1 \end{cases}$$

i
$$\begin{cases} y = 2x^2 - 6x + 10 \\ y = x^2 - 2x + 6 \end{cases}$$

j
$$\begin{cases} y = x^2 - 18 \\ y = 2 - x^2 \end{cases}$$

16 Find the points of intersection for each pair of equations:

a
$$\begin{cases} y = x^2 \\ y = -7x \end{cases}$$

b
$$\begin{cases} y = x^2 + 3x \\ y = -x + 2 \end{cases}$$

c
$$\begin{cases} y = -2x - 1 \\ y = x^2 - 8x + 8 \end{cases}$$

d
$$\begin{cases} y = 2x^2 + 14x \\ y = 16x \end{cases}$$

e
$$\begin{cases} y = x^2 - 8x \\ y = -7 \end{cases}$$

f
$$\begin{cases} y = (x - 3)(x - 2) \\ y = -13x - 10 \end{cases}$$

g
$$\begin{cases} y = -x^2 + 6 \\ x - y = 0 \end{cases}$$

h
$$\begin{cases} y = (x - 3)^2 \\ x - 2y = 3 \end{cases}$$

i
$$\begin{cases} x^2 + y = 5 \\ 2y = 3x^2 - 10 \end{cases}$$

17 Consider the following system of equations:
$$\begin{cases} y = -x^2 + x \\ y = -4 \end{cases}$$

- a** Find the exact values of x of the points of intersection.
- b** Find the points of intersection in exact form.



- 18** Find the points of intersection of the following two equations. Round your answers to two decimal places.

$$\begin{cases} y = -x^2 - 7x \\ x + y + 1 = 0 \end{cases}$$

- 19** Find the coordinates of the points of intersection for each of the following parabola and line pairs:

- a** The parabola $y = -x^2 - 3x + 1$ and the line $y = -3x - 3$.
- b** The parabola $y = x^2 + 7x$ and the line $y = -12$.
- c** The parabola $y = -x^2$ and the line $y = -2x$.
- d** The parabola $y = x(x - 4)$ and the line $y = -3x + 12$.
- e** The parabola $y = x^2 + 6x + 10$ and the line $y = -3x - 10$.

- 20** Consider the parabola $y = x^2 - 5$. Fiona discovered that the line with equation $y = \frac{4x - 21}{4}$ intersects the parabola at exactly one point. Determine the equation of another straight line that has exactly one point of intersection with the parabola, and that has the same y -intercept as $y = \frac{4x - 21}{4}$.

- 21** Consider the following pair of parabolas:

$$\begin{cases} y = 2x^2 + 11x + 1 \\ y = x^2 + 3x - 3 \end{cases}$$

- a** Find the exact value of the x -coordinates of their points of intersection.
 - b** Are the following statements true about these two parabolas?
 - i** They do not intersect each other because one of the parabolas has greater concavity than the other.
 - ii** They are tangents to each other at exactly one point.
 - iii** They have two points in common.
- 22** The parabolas $y = c - ax^2$ and $y = x^2 + 2x + c$ intersect at the point $(2, 0)$.
- a** Solve for c .
 - b** Solve for a .
 - c** Graph the parabolas on the same coordinate plane.

Quadratic equations from systems



- 23** The graphs of the two equations below are graphed and the x -coordinates of their points of intersection are found to be x_1 and x_2 .

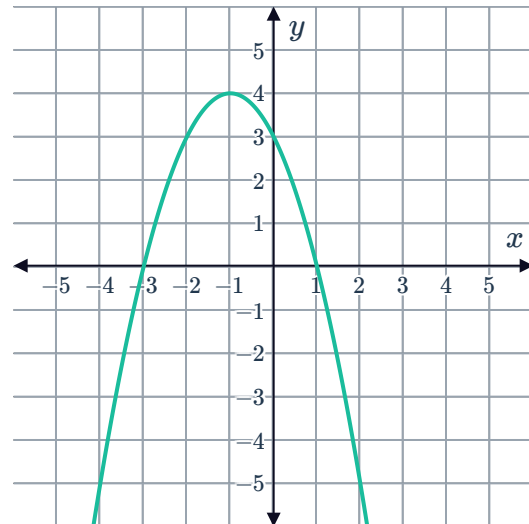
$$\begin{cases} y - 3x^2 - x - 7 = 0 \\ y = 3x - 2 \end{cases}$$

State the quadratic equation whose solutions are x_1 and x_2 . Express the equation in the form $ax^2 + bx + c = 0$, where $a > 0$.

- 24** Consider the quadratic function $y = x^2 + 4$ and the linear function $x + y = 4$.
- a** Sketch the graphs of $y = x^2 + 4$ and $x + y = 4$ on the same coordinate plane.
 - b** Determine the number of solutions to the equation $x^2 + 4 = -x + 4$.
 - c** Find the points of intersection of the two functions.
- 25** If you were given the graph of $y = x^2$, what suitable straight line would you need to graph on the same coordinate plane to solve the equation $x^2 - 2x - 6 = 0$?

- 26** The function $y = -x^2 - 2x + 3$ is shown on the graph:

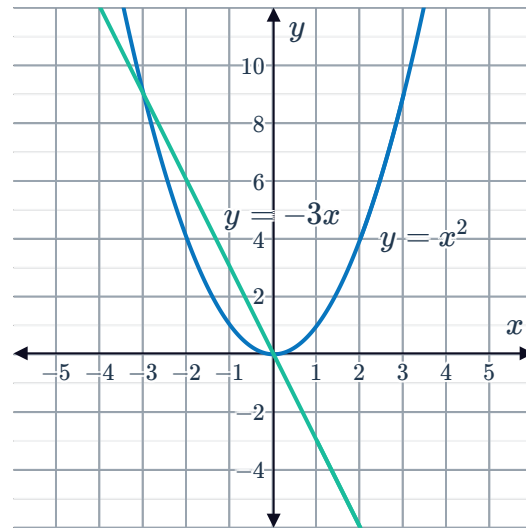
- a** State the linear expression that must be added to $-x^2 + x + 2$ to form $-x^2 - 2x + 3$
- b** Graph the straight line on the same coordinate plane.
- c** Now determine the solutions to $-x^2 + x + 2 = 0$.



- 27 The following functions are displayed on the given graph:



$$\begin{cases} y = x^2 \\ y = -3x \end{cases}$$



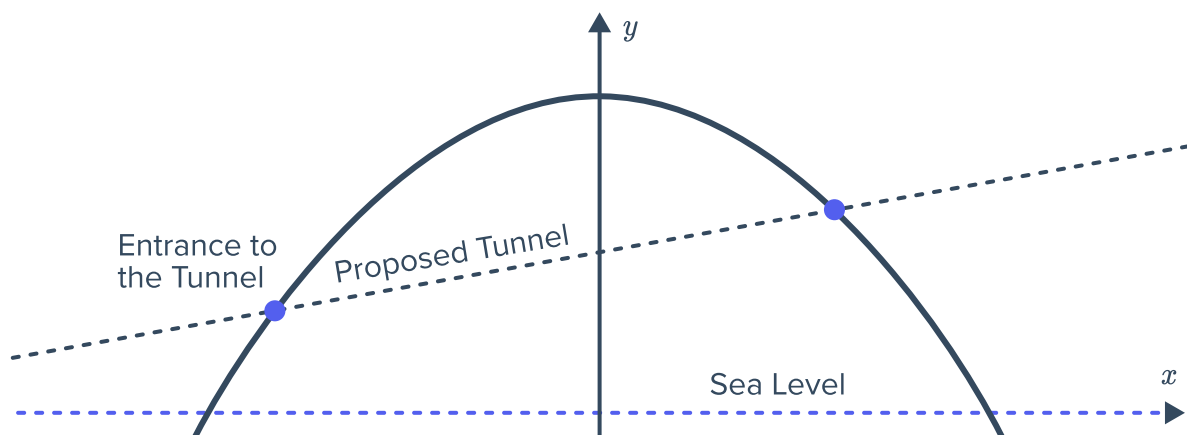
- a How many points on the plane are common to both graphs?
- b State the coordinates of the points of intersection.
- c Now state the solutions to the equation $x^2 = -3x$.

- 28 Consider the functions $y = 9$ and $y = x^2$.

- a Graph the functions on the same coordinate plane.
- b State the coordinates of the points of intersection.
- c Now state the solutions to the equation $x^2 = 9$.

Applications

- 29 The plan for a proposed tunnel through a mountain side is shown in the following diagram:



The height of the mountain above sea level is given by the equation $y = -\frac{x^2}{3} + 4$, and the height of the tunnel above sea level is given by the equation $y = \frac{x}{3} + 2$, where y represents the height above sea level in kilometers.

Solve for x , the heights above sea level at which cars will enter and exit the tunnel



- 30** In business, supply and demand functions are ways of expressing the supply or demand of a commodity as a function of its unit price. Market equilibrium is the term that describes when the supply is equal to the demand. Consider the supply and demand functions for a new power drill below, where y is in dollars and x is in thousands of units supplied.

Supply function

$$y = 4x^2$$

Demand function

$$y = -6x + 5x^2$$

- a** Solve for the value of x , where $x > 0$.
 - b** Find the corresponding value of y .
 - c** Interpret these values of x and y in context.
- 31** Sketch the graph of:

$$h(x) = \begin{cases} 3x - 2, & x < 0 \\ x^2 - 3x - 4, & 0 \leq x < 4 \end{cases}$$